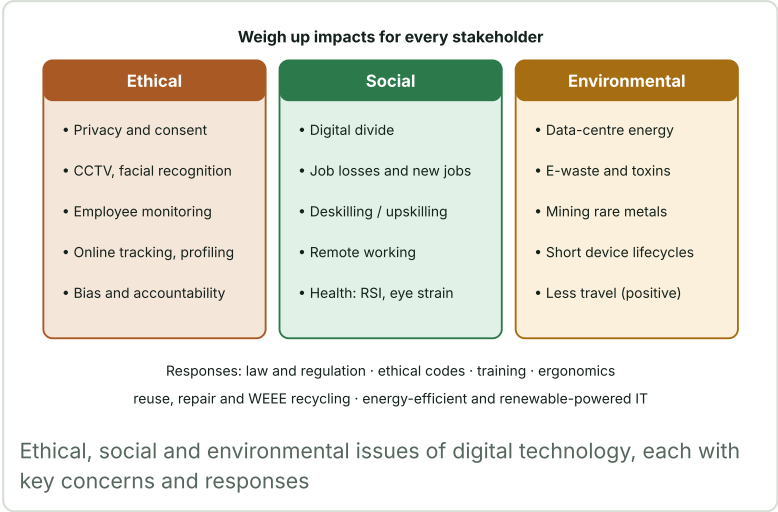
Ethical, social and environmental issues

IS18 FACT FILE

Specification: students should be able to

- Discuss ethical issues including privacy and surveillance
- Discuss social issues including the digital divide, employment and health and safety
- Discuss environmental issues including energy use, sustainability and e-waste

Key words: ethics, privacy, surveillance, digital divide, deskilling, upskilling, remote working, ergonomics, repetitive strain injury, sustainability, e-waste, carbon footprint



Ethical, social and environmental issues of digital technology, each with key concerns and responses

Privacy and surveillance

Digital technology makes it easy to collect, combine and analyse information about people through CCTV, facial recognition, location tracking, online cookies and employee monitoring. Supporters argue this improves security, crime prevention and convenience. Critics point to loss of privacy, chilling effects on free expression, the risk of misuse or data breaches, and inaccurate or biased systems. Surveillance should be lawful, necessary and proportionate, and people should be told what is collected and why.

Social issues

The digital divide separates people who have affordable access to devices, connectivity and digital skills from those who do not, excluding them from services, education and work. Automation removes some routine jobs while creating skilled ones, so retraining is essential. Remote working brings flexibility but can cause isolation. Health risks of prolonged computer use include repetitive strain injury, eye strain, back problems and stress, which are reduced by ergonomic equipment, breaks and sensible policies.

Environmental issues

Producing, powering and disposing of digital technology has environmental costs: mining of scarce materials, the large energy use of data centres and networks, and e-waste containing toxic substances. Short replacement cycles worsen the problem. Organisations can reduce their impact by extending device life, refurbishing and recycling through approved schemes, buying energy-efficient equipment, virtualising servers and using renewable-powered cloud providers. Technology can also help the environment through reduced travel and smarter energy use.

T Key terms from the mark schemes

Digital divide
The gap between people who have access to digital technology and the skills to use it, and those who do not.

Surveillance
The monitoring of people's behaviour, activities or communications, for example by CCTV, tracking or interception.

Sustainability
Meeting present needs without compromising the ability of future generations to meet theirs, for example by reducing energy use and e-waste.

E-waste
Discarded electrical and electronic devices, which may contain toxic and valuable materials.

Repetitive strain injury (RSI)
Pain or damage to muscles, tendons or nerves caused by repeated movements such as typing or using a mouse.

Exam focus

Ethical, social and environmental issues are often tested with extended QWC questions worth 10–15 marks. Plan quickly: identify several distinct issues, explain each with a technical point and an applied example, consider different stakeholders, and weigh positive against negative effects. Use paragraphs and technical vocabulary. A concluding judgement that follows from your arguments is needed for the top band; a list of unexplained points stays in the lowest band however long it is.

? Check yourself

1. Define the term digital divide. [2]
2. Give two causes of the digital divide in Northern Ireland and suggest a way to reduce each. [4]
3. Describe two health problems that can result from prolonged computer use and how each can be prevented. [4]

Answers: 1. The digital divide is the gap between people who have access to digital technology (devices and internet) (1) / and the skills/confidence to use it, and those who do not (1). 2. Two causes (1 each) with a matching solution (1 each), e.g.: poor rural broadband connectivity, e.g. in parts of Fermanagh (1) / government broadband roll-out schemes (e.g. Project Stratum) or satellite/4G-5G alternatives (1) / cost of devices and broadband for low-income households (1) / device lending schemes, subsidised social tariffs (1) / lack of skills or confidence, especially among older people (1) / free digital-skills classes in libraries and community centres (1) / disability (1) / accessible design and assistive technology (1). 3. Two problems (1 each) with prevention (1 each), e.g.: repetitive strain injury (RSI) from repeated keyboard/mouse use (1) / ergonomic keyboards, wrist rests and regular breaks (1) / eye strain and headaches from long periods at screens and glare (1) / anti-glare screens, correct lighting, breaks, free eye tests (1) / back and neck problems from poor posture or unsuitable chairs (1) / adjustable chairs and screens, posture training (1) / stress (1) / workload management, right-to-disconnect policies (1). Accept reference to the Health and Safety at Work (NI) Order 1978 / display screen equipment regulations.